🧱 **BT Bhutan Developer Network (BDN) — Tech Stack (Free + Scalable)**

⚡ **1. Frontend (Core UI Layer)**

✅ **Framework**

- **Next.js (App Router)** → your main engine
- **Tailwind CSS** → fast UI, no design friction

✅ **UI Enhancements**

- **shadcn/ui** → clean, modern components (buttons, cards, dialogs)
- **lucide-react** → icons

👉 Why this works:

- Fast to build
- Looks premium without design overhead
- Industry standard

🧠 **2. Backend (No Traditional Backend Needed)**

✅ **Option A (Recommended): Supabase (ALL-IN-ONE)**

Use:

- **Supabase Auth** → Google + Email login
- **Supabase Postgres DB** → users, posts, teams
- **Supabase Realtime** → live feed updates
- **Supabase Storage** → profile images

👉 This replaces:

- Node backend ❌
- Firebase complexity ❌

🔁 **Alternative (if you prefer Firebase)**

- Firebase Auth
- Firestore DB
- Firebase Storage

👉 But Supabase is better for:

- SQL (you understand structure better)

- scalability
- clean schema control

🗄️ 3. Database Schema (Simple & Powerful)

Use PostgreSQL (via Supabase)

Core Tables:

👤 users

- id
- name
- email
- bio
- skills (array or text)
- github_url
- availability
- is_verified (Bhutan check)

📝 posts

- id
- user_id
- type (help / team / project)
- content
- created_at

💬 comments

- id
- post_id
- user_id
- content

🤝 teams (optional later)

- id

- title
- description
- created_by

👉 Keep schema minimal. You can expand later.

🔐 **4. Authentication**

Use:

- **Supabase Auth**

Enable:

- Google login
- Email login

👉 Later:

- Add **Bhutan verification layer**
    - manual approval OR
    - restrict domain (e.g. college/work emails)

🌐 **5. Hosting (100% Free)**

Frontend

- **Vercel**
    - Auto deploy from GitHub
    - Perfect for Next.js

Backend (Supabase)

- Hosted automatically

👉 Zero DevOps needed

⚡ **6. Real-Time Features (Important)**

Use:

- **Supabase Realtime**

For:

- live posts

- comments updating instantly

👉 Makes your platform feel alive

🧩 7. Core Features → Tech Mapping

Feature	Tech
Login	Supabase Auth
Profiles	Supabase DB
Feed	Supabase + Next.js
Comments	Supabase Realtime
Team posts	Same posts table (type = "team")
Notifications (later)	Supabase or simple polling

🎨 8. UI Structure (Pages)

Public

- / → Landing page
- /explore → view community posts

Authenticated

- /dashboard → main feed
- /profile → user profile
- /create-post
- /teams (later)

⚙️ 9. Folder Structure (Clean Setup)

```
/app
  /dashboard
  /profile
  /create
/components
  Navbar.tsx
  PostCard.tsx
  ProfileCard.tsx
/lib
  supabaseClient.ts
```

🔥 10. MVP Build Plan (Realistic)

Week 1 (Core Build)

- Auth (Google login)
- User profile creation
- Post creation
- Feed display

Week 2 (Polish)

- Comments
- UI improvements
- basic Bhutan filtering

🚀 11. Scaling Path (Future)

After traction:

- Add:
  - Reputation system
  - Verified developers
  - Hiring posts
  - Startup co-founder matching
- Upgrade:
  - Edge functions (Supabase)
  - Notification system

💡 Final Architecture (Simple View)

[Next.js Frontend]
↓
[Supabase (Auth + DB + Realtime + Storage)]
↓
[Vercel Hosting]

👉 Clean. Fast. Scalable.